

Question 1.1. At least _____% of the people are between 25 and 65 years old. Explain your answer!
(6 Points)

Trước tiên, tính khoảng cách từ giá trị trung bình đến các giới hạn:

Khoảng cách từ 45 đến 25 = 20 năm = 4 Khoảng cách từ 45 đến 65 = 20 năm = 4 Theo bất đẳng thức Chebyshev, xác suất mà một giá trị nằm ngoài khoảng $[-k, +k]$ không quá $1/k^2$.

Ở đây, $k = 4$, nên xác suất nằm ngoài khoảng $[45 - 20, 45 + 20] = [25, 65]$ không quá $1/4^2 = 1/16 = 0.0625 = 6.25\%$

Do đó, xác suất nằm trong khoảng $[25, 65]$ ít nhất là $1 - 0.0625 = 0.9375 = 93.75\%$

-> 93.75

Question 1.2. At most _____% of the people have ages that are not in the range 25 years to 65 years.
Explain your answer! **(6 Points)**

6.25

Question 1.3. At most _____% of the people are more than 65 years old. Explain your answer! (6 Points)

Hint: If you're stuck, try thinking about what the distribution may look like in this case.

Đối với câu hỏi này, chúng ta cần áp dụng bất đẳng thức Chebyshev một phía (one-sided Chebyshev), còn được gọi là bất đẳng thức Cantelli:

Với bất kỳ biến ngẫu nhiên X có trung bình μ và độ lệch chuẩn σ : $P(X - \mu \geq k\sigma) \leq 1/(1 + k^2)$ Áp dụng với $k = 4$ (vì $65 = 45 + 4 \times 5$):

$$P(X \geq 65) = P(X - 45 \geq 20) \leq 1/(1 + 4^2) = 1/17 \approx 0.0588$$

5.88

Question 2.2. Suppose the data science class decides to construct a 90% confidence interval instead of a 95% confidence interval, but they still require that the width of the interval is no more than 6% from left end to right end. Will they need the same sample size as in 2.1? Pick the right answer and explain further without calculation. **(6 Points)**

1. Yes, they must use the same sample size.
2. No, a smaller sample size will work.
3. No, they will need a bigger sample.

Question 2.4. This shows that the percentage in a normal distribution that is at most 1.65 SDs above average is about **95%**. Explain why 1.65 is the right number of SDs to use when constructing a **90%** confidence interval. **(6 Points)**

The function `stats.norm.cdf(1.65)` calculates the proportion of values below 1.65 standard deviations on a normal distribution. In contrast, a confidence interval represents the central portion of the distribution, excluding the extreme values at both tails.

Question 3.2. Why does the Central Limit Theorem (CLT) apply in this situation, and how does it explain the distribution we see above? We recommend reviewing [14.4](#) for a refresher on CLT. **(6 points)**

Type your answer here, replacing this text.

